Modeling of synchronous generators in power system studies

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Abstract:

Round-rotor synchronous generators, used in steam- and gas-turbine power plants, have historically been modeled in stability studies using the popular *genrou* model, a model available in all commercial power system simulation programs. Salient-pole synchronous generators, most typically used in hydro power plants, have historically been modeled using the *gensal* model, also available in all commercial power system simulation programs. For round-rotor machines, the Western Electricity Coordinating Council (WECC) has for decades also used an alternative model called *gentpf*. Recently, WECC started to discontinue the use of the *gensal* model for salient pole machines and instead uses a modified version of the *gentpf* model called *gentpj*. This paper presents clear evidence, from measurements of on-line response of large generating units, showing the significantly better performance of the *gentpf* and *gentpj* models as compared to *genrou* and *gensal*. Finally, this paper documents clearly the relationship of these models to each other and to the general synchronous machine model, which thus offers some insight into the differences among them.

1. INTRODUCTION

Computer-simulation models for the purpose of analyzing the dynamic performance of synchronous generators is a well-established practice and has been a core part of power system analysis for decades. The development of mathematical models for the simulation of the dynamic behavior of synchronous machines dates back to the late 1920's with the original papers by Park [1]. Two of the classic and definitive text books on the subject include [2] and [3], and more recent references such as [4] and [5] contain comprehensive treatment of the subject of synchronous generator modeling for use in computer simulation programs. The incorporation of such models into large scale power system simulations was first developed in the late 1960's [6].

The most commonly used synchronous generator models in the power industry, over the past several decades, for representing large round-rotor synchronous generators is the so-called *genrou* model, available in numerous commercial software platforms. For salient-pole machines the most common model is *gensal*. These two models are used by countless utilities in North America and world-wide. For round-rotor machines, in the Western Electricity Coordination Council (WECC), another model commonly used is the so-called *gentpf* model, which is based on the model described in [7]. In recent years, however, WECC has adopted a slightly modified version of this latter model, now called the *gentpj* model, for the simulation of salient-pole synchronous generators used in hydro power plants [8] and [9]. Evidence in WECC has shown that the *gensal* model is unable to properly capture the steady-state behavior of the machines [9]. In this paper this will be formally documented.

This paper offers the following contributions and insight. Although the *gentpf* model has been in use for many years in WECC, the authors could not find any documentation or evidence in the literature that shows its relationship to *genrou*, nor any benefits it might offer in comparison to *genrou*. Recent work has provided significant insight and in this paper that evidence is provided and explained to show that the *gentpf* model gives a significantly better

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KEYWORDS

Power Generation Modeling, Synchronous Generator Modeling, Model Validation

representation of the field current response of the round-rotor synchronous generator in many cases. Furthermore, the *gentpj* model is explained and example cases provided to show evidence from field measurements where this model may be needed for large round-rotor machines. Finally, field evidence is also provided to show the superiority of the *gentpj* model when modeling large salient-pole synchronous generators associated with hydro power plants. The appendix of the paper also provides clear documentation of the two models. Thus, *gentpj* may be deemed to be the most general of the models, and when appropriately parameterized (explained in the appendix and section 3) can accurately represent the behavior of both round-rotor and salient-pole machines.

The remainder of this paper is organized as follows. Section 2 provides a brief account of the relationship of the four models *genrou, gensal, gentpf* and *gentpj*. This is complemented by the appendix, which includes the full equations for the lesser known *gentpf* and *gentpj* models. Section 3 provides field evidence from on-line measured response of synchronous machines both in steady-state and transients, comparing the performance of the various models for synchronous machines. This includes two generators from fossil fuel plants and one large nuclear unit, as case studies for round-rotor machines. The last cases study is of a large salient-pole machines in a hydro power plant, comparing the performance of the *gensal* and *gentpj* models. Section 4 finally concludes the paper by summarizing the results, conclusions and recommendations of the paper.

2. The synchronous generator models

The *genrou* and *gensal* models have been used for decades in numerous commercial simulation programs for modeling round-rotor and salient-pole synchronous generators. A detailed explanation of these models may be found in numerous manuals of commercial tools, such as [10]. There is no published documentation that the authors could find, on the *gentpf* model, although it has been used for decades, and was developed in WECC. The appendix provides the equations for this model. The *gentpj* model was developed by J. M. Undrill, D. N. Kosterev and H. Yang [8]. The only difference between the *gentpf* and *gentpj* models is the introduction of one extra parameter, K_{is}, in the latter model – the parameter is explained later in this paper.

It is pertinent to provide here an explanation of the differences among these models. Table 1 shows the parameters associated with these four generator models. All of the listed parameters, with the exception of K_{is}, are well understood and documented in countless references, e.g. [3], [4] and [5].

The parameters listed in Table 1, with the exception of K_{is}, all are readily provided by the original equipment manufacturer (OEM) for all synchronous generators manufactured in the last few decades. For older units, and in some rare cases, all of these parameters may not be available. In such cases, techniques such as presented in [16] may be used to estimate the unavailable parameters. However, where possible, and certainly for modern units from major OEMs, the complete parameters should be requested from the OEM. As noted in [3], in the general model of a synchronous machine there is a single winding on the d-axis which represents the field winding. There is, however, an infinite number of possible paths for eddy currents to flow in closed circuits on the rotor. To properly capture the effect of all these eddy current loops, countless damper windings should be modelled on the direct (d) and quadrature (q) axes of the Park transformer general machine model. This of course is not practical. For machine design most OEMs will use finite-element models. While for power system studies, it has been customary to assume one damper winding on the d-axis and one damper winding on the q-axis [3]. This leads to the most commonly used general model of a synchronous machine, shown pictorially in Figure 1. The generalized model of a synchronous generator is a couple-circuit model, two circuits of inductances and resistances representing each of the two axes – the d-axis and q-axis. In practice, however, it is difficult at best to directly measure the individual circuit parameters (self and mutual inductances) and so the common practice from early days has been to define impedances and time-constants (the parameters in Table 1) that can be measured from standardized tests [12], [13]. In this way the coupled-circuit representation is translated into an operation-impedance model that constitutes the impedances and time-constants listed in Table 1. It can be shown that one can convert the operation-impedance parameters to the couple-circuit parameters (see Table

4.11 in reference [11]). All of the models being discussed in this paper come from this general model, after certain further simplifying assumptions.

It is pertinent to summarize some of the key assumptions that lead to each of these models, as this leads to some key insights, which cannot be found elsewhere in one concise treatment. There are a set of simplifying assumptions that are common to all four of the models, e.g. that the armature currents do not produce any harmonic magnetomotive forces [3]. These common assumptions will not be repeated and can be found in the standard literature [3], [4] and [5].

Table 1: Parameters of the four generator models.

Parameter	genrou	gensal	gentpf	gentpj
Xd	√	√	√	√
Xq	√	√	√	√
X'd	√	√	√	√
X'q	√		√	√
X''d	√	√	√	√
X''q	=X''d	=X''d	√	√
Xl	√	√	√	√
T'do	√	√	√	√
T'qo	√		√	√
T''do	√	√	√	√
T''qo	√	√	√	√
S10	√	√	√	√
S12	√	√	√	√
Kis				√

As an example, let us consider one of the five differential equations associated with the general machine model in Figure 1, the equation for the d-axis damper winding.

$$e_{kd} = i_{kd}r_{kd} + si_{kd}L_{kd} + si_fL_{fkd} + si_dL_{akd} \qquad (1)$$

where *Lkd* is the total self-inductance of the d-axis damper (sum of the mutual-inductance and the winding's leakage inductance), *Lfkd* is the mutual-inductance between the field and damper winding, *Lakd* is the mutual between the stator d-axis and the damper winding, and the operator s is *d/dt*. The voltage across the winding is *ekd, rkd* is the resistance of the winding and the *i** are the various winding currents.

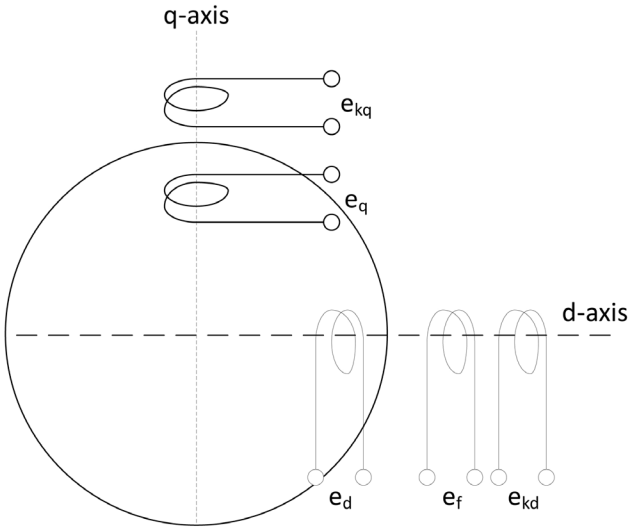


Figure 1: General synchronous machine model in Park's reference frame.

First consider *genrou*. In this model three key assumptions are made: (i) rotor saliency is ignored in the subtransient time frame, that is *X"q* is set equal to *X"d*, (ii) all the mutual-inductances between the windings on the same axis are assumed to be equal in deriving the operation impedance model (i.e. on the d-axis *Lmd = Lfkd = Laf = Lakd* and on the q-axis *Lmq = Lakq*), and (iii) saturation is modeled using the open-circuit saturation curve (fit to a quadratic model[5]); furthermore, it is assumed that all the inductances saturate in the same way and that saturation is not a function of machine loading. These assumptions when applied to the general model lead to the *genrou* model, shown in Figure 2. Also, note that saturation is a function of the voltage behind the subtransient reactance *(X"d)*.

To go from the *genrou* to *gensal*, two additional assumptions are made: (i) since the q-axis has significantly less iron in a salient-pole machine, then transiency can be ignored (eliminated *X'q* and *T'qo*), and (ii) saturation is modelled as in *genrou*, but applied only to the d-axis. Also, it is typical for the gensal model to assume *X"q = X"d*. Thus, the d-axis circuit/block diagram of the gensal model is essentially identical to *genrou*, while the q-axis

[5]- Another model available in Siemens PTI PSS*E, genroe, is essentially identical to genrou, except that the saturation model is fit by an exponential, rather than quadratic, equation.

model is significantly simplified by eliminating transiency – see Figure 3. The one difference is that saturation is now a function of the voltage behind the transient reactance *(X'd)*.

To get the *gentpf* model, however, we start with a different set of assumptions, namely (i) rotor saliency in the subtransient time frame is not ignored (i.e. *X"q* is not necessarily set to be equal to *X"d*, although it can be depending on the input data), (ii) the following relationship is assumed, between the self and mutual inductances on the d-axis [7],

$$\frac{L_{akd}L_{fkd}}{L_{kd}L_{af}} = 1 \qquad (2)$$

and (iii) saturation is again quadratic and based on the open-circuit saturation curve of the machine, but applied to the voltage behind leakage reactance. These assumptions, after much algebraic manipulation, lead to *gentpf* (Figure 4). Note in reference [7] all these three assumptions are not made, and so the model presented there is not exactly *gentpf*. In reference [7] D. W. Olive states that the assumed relationship in equation (2) essentially corresponds to the assumption that "any transient change in stator currents will be reflected initially in the damper windings and will not affect the field."

Finally, one small addition leads us from *gentpf* to the *gentpj*, namely, the introduction of a new parameter K_{is}. Referring to Figure 4, the saturation coefficients Sd and Sq in *gentpf* are given by:

$$Sd = f_s(E_l) \text{ and } Sq = \frac{X_q}{X_d} f_s(E_l) \qquad (3)$$

where *fs*( ) is a quadratic function that is defined by the two saturation parameters *S10* and *S12*, which are derived from the open-circuit saturation curve of the machine [4], and *El* is the voltage behind the leakage reactance of the machine. One of the key simplifying assumptions in all of the generator models is that the saturation of the machine under loaded conditions is the same as under no-load conditions, and thus the measured open-circuit saturation curve of the machine is used as stated above in the generator model over the entire operating range of the machine. A detailed treatment is provided in [5] that shows this is not in fact true for most large synchronous generators, which is not surprising since as the machine is loaded and stator current increases this will no doubt affect the magnetic flux in the rotor circuits and thus increase the level of saturation. Some of the references discussed in this treatment in [5] employ significantly more complex models to capture this and other phenomena neglected in the typical models used in stability studies. Such models are rather complex for large scale power system simulations. In *gentpj* a simple modification was proposed by J. M. Undrill [8], which carries almost no additional computation burden, to capture the load dependency of saturation. The saturation function is changed from equation (3) to

$$Sd = f_s(E_l + K_{is}I) \text{ and } Sq = \frac{X_q}{X_d} f_s(E_l + K_{is}I) \qquad (4)$$

where *I* is the magnitude of the stator current, and K_{is} is a constant. In the next section we will show the significance of these different modeling assumptions when applying the models to simulate actual measured generator performance.

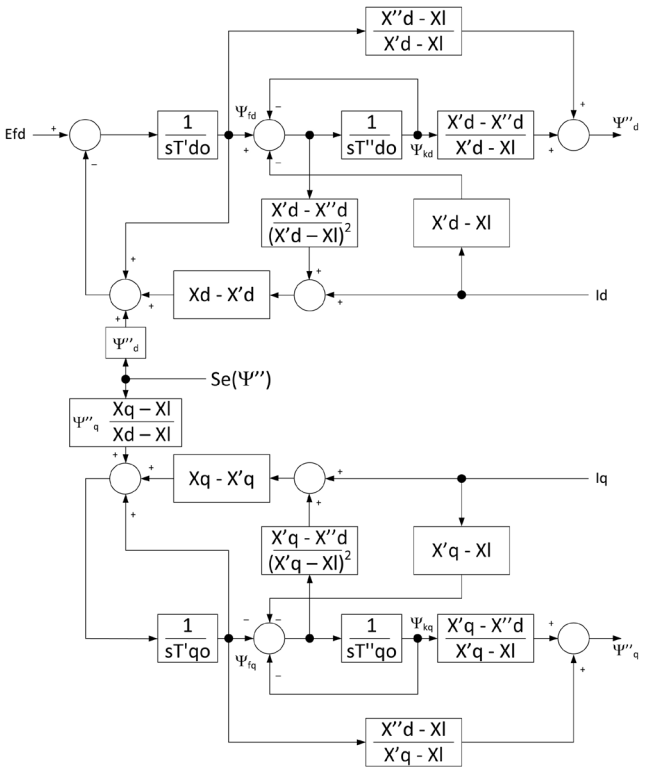


Figure 2: The *genrou* model. The sign convention used here is similar to that used in the GE PSLF™ program, which is different to that used in some other commercial tools (e.g. Siemens PTI PSS®E and PowerWorld Simulator). Both sign conventions are valid.

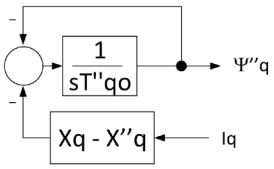


Figure 3: The q-axis model of gensal. The d-axis is identical to *genrou*, except that the saturation function acts on the voltage behind X'd.

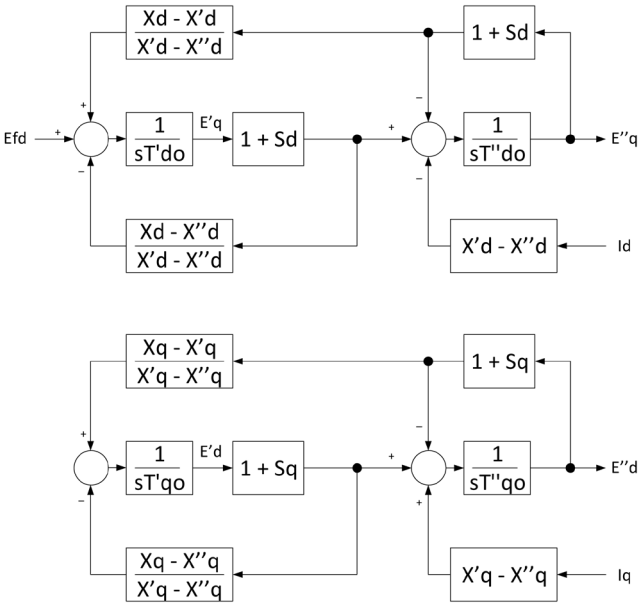


Figure 4: The *gentpf* generator model.

3. Comparison of the performance of the four different generator models based on measured on-line response of synchronous machines

3.1. Case Study 1 – Round-Rotor Generator, Steam-Turbine

The first example is of a 240 MVA round-rotor generator driven by a steam-turbine. On-line voltage reference step tests were done on the unit while operating in-service at 160 MW. A set of steady-state measurements were taken for a range of reactive power outputs at this loading level, monotonically, starting with leading power factor and gradually going towards lagging power factor. That is, a V-curve was recorded. Using the OEM data for the generator (i.e. *Xd, X'd, X''d*, etc.) and the *genrou* model, the comparison of the measured and simulated results for both the V-curve and the step-test are shown in Figures 5 and 6. Then the exact same data was used, this time with the *gentpf* model. In both cases the validated exciter model is used; the exciter model was validated using the tool and technique presented in [14]. These results are also shown in Figure 5 and 6. Thus, the only difference between the two simulations is the generator model *genrou* versus *gentpf* – all the other models and parameters are identical, even the parameters for the generator model. In steady-state both cases provide a relatively good match between measured and simulated response with a maximum error of 0.8% in the field current estimated compared to measurement – this can easily be attributed to measurement error. For the step-test, however, *gentpf* clearly gives a superior match for the field current response. In the step-test the error in the field current transient response, on the first peak, is 15% for *genrou*, as compared to just under 4% for *gentpf*.

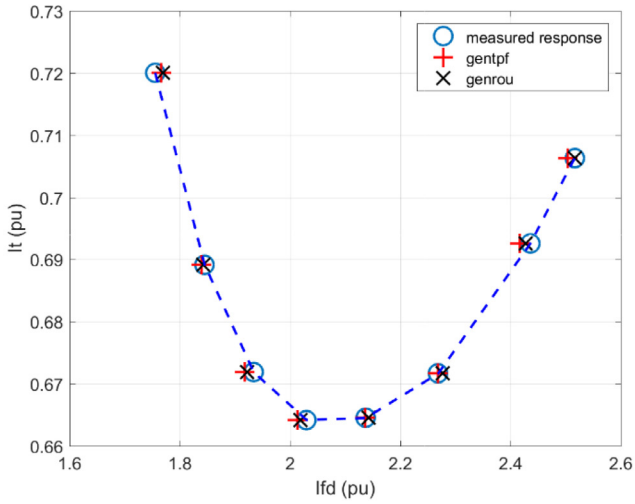


Figure 5: V-curve measured for case 1. It – stator current and Ifd – field current. Both values are in per unit (pu).

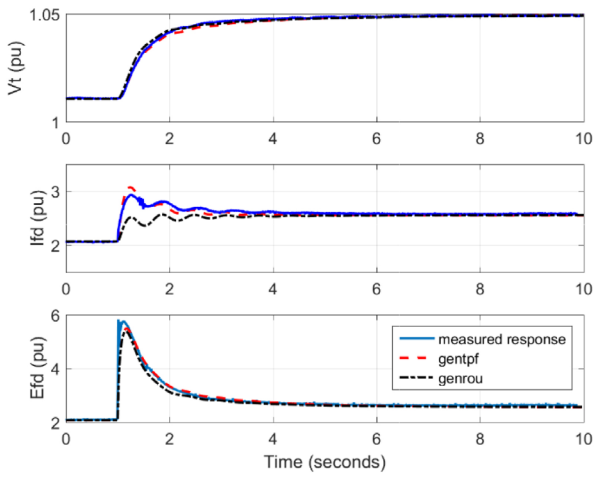


Figure 6: On-line voltage reference step test for case 1. Stator voltage (Vt), field current (Ifd) and field voltage (Vfd) in per unit (pu).

3.2. Case Study 2 – Round-Rotor Generator, Steam-Turbine

The second case is for a 618 MVA round-rotor machine, also driven by a steam-turbine. In this case on-line and off-line voltage reference step-tests were performed. Once again the OEM data was used with both the *genrou* and this time *gentpj* models. The exciter model was validated again using the tool and technique in [14], and the exact same exciter model and parameters used in all cases. The step-tests were done at 0 MW (off-line), 200 MW and 500 MW loading on the machine. The results are shown in Figures 7, 8 and 9, respectively. It is clear than the *gentpj* model is superior in matching the field current response over all the operating conditions. The K_{is} parameter was not provided in the OEM data and is the only parameter that is different between the two generator models. It was estimated using the tool described in [14], and the measured data. That is, using a least-square estimation technique and the steady-state equation for the model to fit the model to the measured V-curve data, K_{is} can be easily determined. It is also interesting to note that *gentpf* does as good a job as *gentpj* for the off-line and partial-load case, but for the near baseload case it incurs an offset error. This is clearly due to the changing saturation level of the machine with increased loading, which is captured by the K_{is} parameter in *gentpj*.

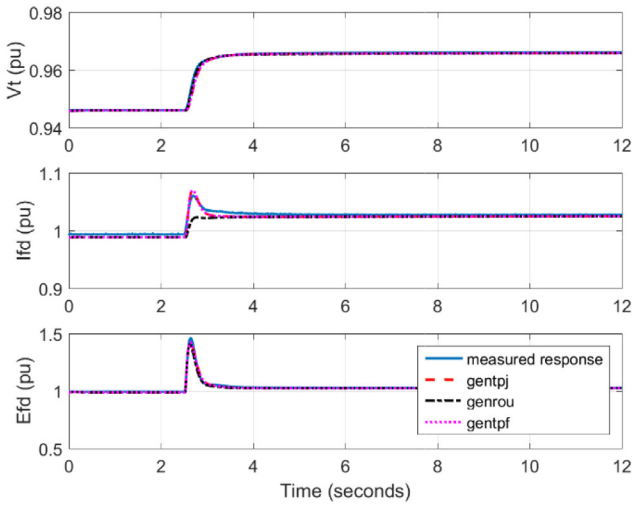


Figure 7: Off-line voltage reference step test for case 2.

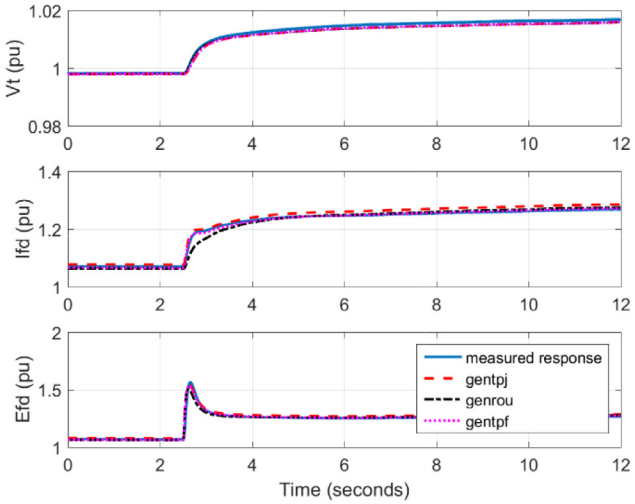


Figure 8: On-line voltage reference step test for case 2; unit at 120 MW.

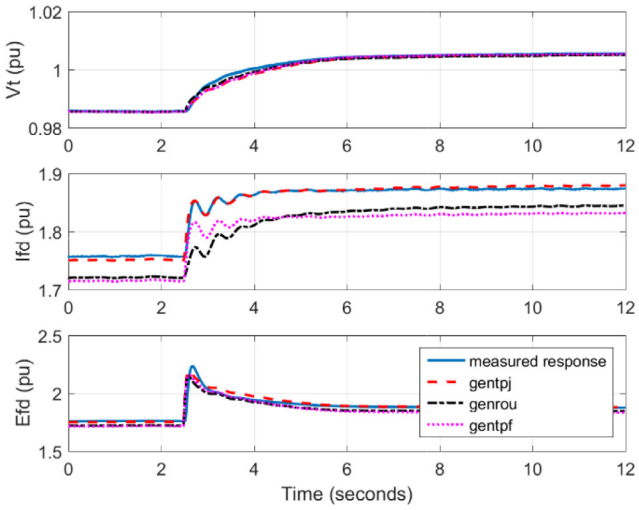


Figure 9: On-line voltage reference step test for case 2; unit at 500 MW.

3.3. Case Study 3 – Round-Rotor Generator, Nuclear Unit

The third case is for a 1559 MVA round-rotor machine, also driven by a steam-turbine in a nuclear power plant. In this case we show one of the on-line voltage reference step-tests and a V-curve, both measured at 1350 MW, which essentially baseload on the unit. In this case three models were investigated:

- *genrou* with the OEM data for all the parameters in Table 1,

- *genrou* with the OEM data, but three parameters were adjusted per previous validation work [15] – this is called the *genrou adjusted* model, and
- *gentpj* with all the OEM data and the K_{is} parameter fitted based on the V-curve data.

In 2004 it had been observed, from a system event, that the units at the plant did not correctly represent the actual reactive response observed in the field at baseload [15]. As such, since the *gentpj* model was not yet developed at that time, adjustments were made to *Xd, S10* and *S12* (by as much as 20%) to achieve a better match. Here this *adjusted genrou* model is also used, where all but these three parameters are the same as the OEM provided values. In all cases, the same (validated) exciter model is used. The results are shown in Figures 10 and 11.

Once again we see the same performance, the *gentpj* model is superior to the *genrou* model, when using the OEM data, in fitting the field current response both in steady-state and dynamically. It is interesting to note that the *adjusted genrou* model also performs as well as the *gentpj* model at baseload. However, based on the results of case 2, it is perhaps reasonable to believe that the *gentpj* model is likely to be a better match for the unit over a wider range of operating conditions. Given that this is a nuclear unit, it was not possible to do partial and no-load V-curve and step-tests to confirm this expectation, since operation at partial and no-load conditions would be both costly (lost opportunity cost) and require significant additional approvals and engineering review.

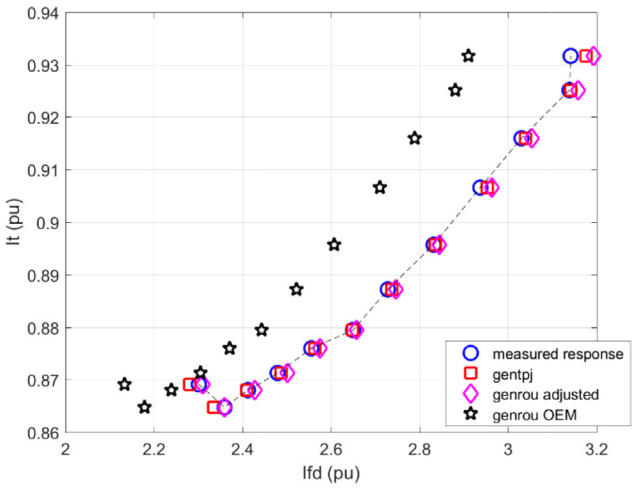


Figure 10: V-curve for case 3 measured at near baseload.

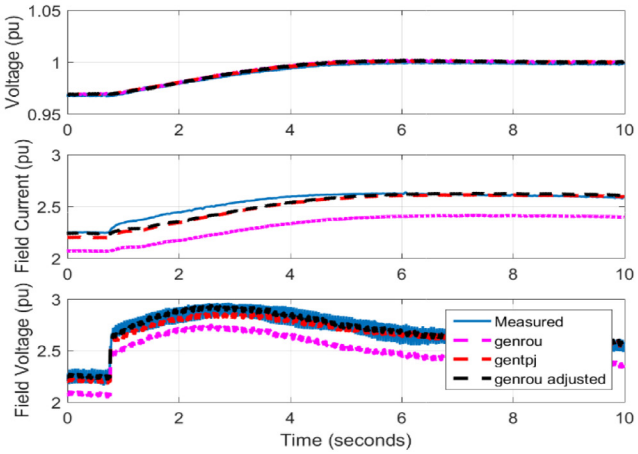


Figure 11: On-line voltage reference step test for case 3 measured at near baseload.

3.4. Case Study 4 – Salient-Pole Generator, Hydro-Turbine

The final case is one example of tests on an 80 MW hydro unit with a salient-pole synchronous generator. In this case the *gensal* model performance was compared to *gentpj*. For a salient-pole machine it is reasonable to assume that the quadrature axis has no transiency due to the lack of iron in the q-axis path as compared to the d-axis. Therefore, in the case of using the *gentpj* model, *X'q* is set equal to *Xq* and *T'qo* is set to an extremely large value. This essentially has the effect of reducing the q-axis model of the *gentpj* model to that shown in Figure 12. Comparing the Figure 12 to Figure 2, the main difference between the q-axis model of the *gensal* and *gentpj* model (when setting the parameters of *gentpj* appropriately for use with a salient-pole machine) is the fact that saturation is modeled in *gentpj* for the q-axis, where as it is ignored in *gensal*. Furthermore, both on the d- and q-axis, saturation in *gentpj* includes the K_{is} factor which models the dependency of saturation on machine loading.

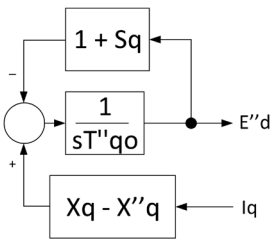


Figure 12: Q-axis model of the *gentpj* model when applying the model for a salient-pole machine by setting Xq=Xq and T'qo = 9999 (i.e. removing the transient circuit E'q).

The results for this case are shown in Figures 13 and 14. Once again we see that the *gentpj* model performs

significantly better. In this case the primary advantage of the *gentpj* model is the ability to more accurately model the steady state value of field current over the entire operating range of the machine. The shape of the transient response of field current is the same in both cases.

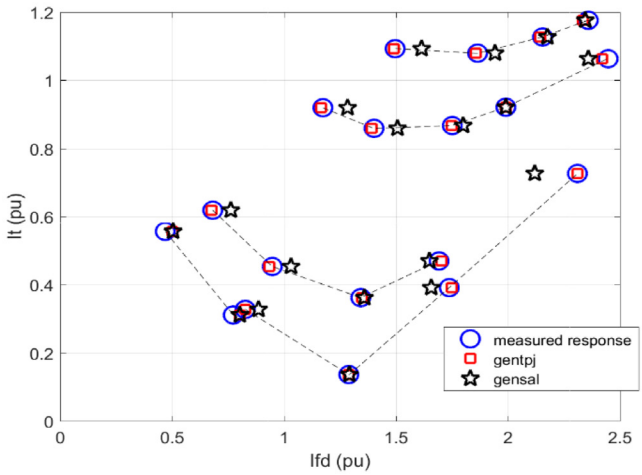


Figure 13: V-curves for test case 4.

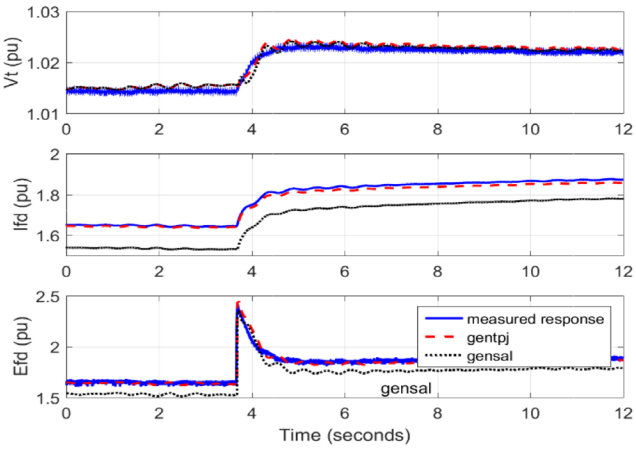


Figure 14: On-line voltage reference step test for case 4; measured at baseload.

4. The importance of proper modeling of field current/voltage response

The question may be asked as to the importance of the differences observed in the performance of the models. In all cases the stator voltage response seems very similar, and the real and reactive power response, though not shown, are identical in all cases, since we are applying the technique presented in [14]. So why should the differences observed in the field current and voltage matter?

The key is that the *genrou* and *gensal* models invariably show a lower steady-state value of field current and voltage, for the same operating condition – i.e. for a given value of stator voltage, real and reactive power output. This result is significant because in planning studies, the *genrou (gensal)* model will give a lower field current value, as compared to the *gentpj* model, for the same response in a simulation. This would mean that in mid-term simulations, with the *genrou (gensal)* model the over-excitation limiter (OEL) model would pick up for a higher value of machine reactive power output, thereby overestimating reactive margins on the system. This is what was reported and observed in [15], which prompted the alterations in the OEM parameters to better match the actual observed response of the nuclear units due to the action of the OEL. This can be illustrated with a very simple simulation. Figure 15 shows the simulated response of one of the units at the nuclear power plant in case 3, when simulating an event that depresses the voltage at the extra high-voltage (EHV) transmission level near the plant. The simulation was performed in a full planning model of the entire interconnection. Clearly, the *genrou* model under estimated the actual field current response, as compared to the *gentpj* model. Note, the same would be true also for under-excited conditions, as can be seen from the V-curves in for example, Figure 13. That is, for the same under-excited level of field current the *gensal* or *genrou* model will over-estimate, at a given real power level and stator voltage, the amount of reactive power absorbed by the machine.

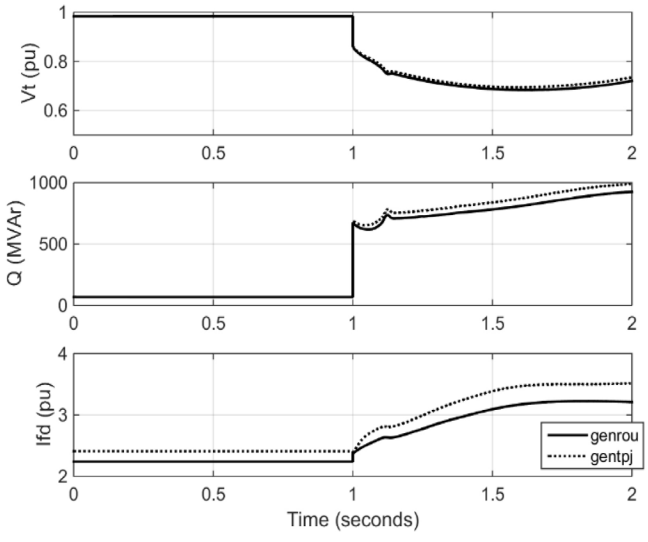


Figure 15: Interconnection wide simulation for the large nuclear unit using either the *genrou* or *gentpj* model (with OEM data) for an event leading to depressed EHV voltages.

5. Conclusions and recommendations

In this paper we have examined the performance of four synchronous generator models currently available in many commercial power system simulation tools, by comparing simulations to actual measured on-line response of large round-rotor and salient-pole synchronous generators. Two of these models, *genrou* and *gensal*, are popular models that have been available and used for decades by the majority of utilities in North America and many other regions for modeling round-rotor and salient-pole generators, respectively. The other two models, *gentpf* and *gentpj*, were developed in the Western Electricity Coordinating Council (WECC), with *gentpj* being developed as recently as 2007 [8]. In the body of the paper, the relationship of these models is described, with respect to the generalized machine model.

Based on the results presented, the following conclusions may be drawn:

- The *gentpj* model certainly provides superior performance in capturing both the transient and steady-state field current response of synchronous generators over the operating range of the machine. The evidence shown proves that this is true for both round-rotor and salient-pole machines. Among them, the authors have numerous other examples to provide similar results as shown in this paper. For the sake of brevity and completeness only four cases were presented.
- The machine parameters used for the *gentpj* model are the same parameters as those used in the *genrou* and *gensal* models, as provided by the OEM on the standard OEM datasheets. The only parameter not provided by the OEM, which is unique to *gentpj*, is K_{is}. This parameter can be easily derived from steady-state V-curve data measured in the field.
- The error in matching the field current response of the generator, as observed for the *genrou* model is twofold. The *genrou* model is unable to capture the increasing effect of saturation with increased machine loading, which subsequently results in an under estimation of the steady-state value of field current (and voltage) required at high loading levels of the machine. This is captured by the K_{is} parameter in *gentpj*. Secondly, in some cases (e.g. case 1 and 2 in this paper) the transient response of the field current is more accurately captured by the *gentpf* and *gentpj* models. This seems to indicate that the underlying assumptions in developing the *gentpf* model (which is also the core of the *gentpj* model) is more indicative of actual machine performance in these cases. We believe the fact that saturation is applied more directly, as a multiplicative term, to all the inductances in the *gentpf* and *gentpj* model, rather than a simple additive term in *genrou*, is the main contributing factor to this result. This second aspect is not necessarily always true (e.g. case 3 in this paper), but even in those cases *gentpf/gentpj* still faithfully capture the transient response of field current.

Based on these results, the decision by WECC to discontinue the use of the *gensal* model for salient-pole machines is quite warranted. Furthermore, it is perhaps appropriate moving forward to consider the use of the *gentpf* model (i.e. *gentpj* with K_{is} = 0) instead of the *genrou* model for round-rotor machines. Where actual on-line measured V-curve data is available, K_{is} should be estimated and the full *gentpj* model used. The parameter K_{is} is constant throughout the range of the generator's operation, since the change in the factor that goes into the saturation equation $(K_{is}.I)$ is nearly proportional to the loading of the machine (i.e. magnitude of stator current). It must be realized that obtaining measured on-line V-curve data may not necessarily always be a simple task for all units (e.g. nuclear units, or peaking units, etc.) and so this should neither be mandated nor an absolute requirement, but rather a goal to be achieved if and when possible and feasible. Finally, OEMs should consider in the future estimating and providing the K_{is} parameter.

6. Appendix

the authors wish to first, and foremost, acknowledge that the *gentpf* model is a model based on [7] and originally developed and used in WECC. Furthermore, the *gentpj* model was originally proposed by J. M. Undrill, as documented in [8] – *gentpj* is a modification of *gentpf*. For the sake of completeness, we provide the equations for these models here, though in a slightly different format so that they are more readily implementable in a software platform. The full model block diagram is also provided in Figure 4.

$$\frac{dE'_q}{dt} = [E_{fd} - (1 + S_d)E_{q1}]\frac{1}{T'_{do}} \quad (5)$$

$$\frac{dE'_d}{dt} = -(1 + S_q)\frac{E_{d1}}{T'_{qo}} \quad (6)$$

$$\frac{dE''_q}{dt} = -(1 + S_d)\left(\frac{X'_d - X''_d}{X_d - X''_d}\right)\frac{E_{q2}}{T''_{do}} \quad (7)$$

$$\frac{dE''_d}{dt} = -(1 + S_q)\left(\frac{X'_q - X''_q}{X_q - X''_q}\right)\frac{E_{d2}}{T''_{qo}} \quad (8)$$

$$V_q = E_{q1} + E_{q2} - I_q r_a - I_d\left(\frac{X_d - X_l}{1 + S_d} + X_l\right) \quad (9)$$

$$V_d = E_{d1} + E_{d2} - I_d r_a + I_q\left(\frac{X_q - X_l}{1 + S_q} + X_l\right) \quad (10)$$

$$E_{q1} = E''_q - E_{q2} + I_d\left(\frac{X_d - X''_d}{1 + S_d}\right) \quad (11)$$

$$E_{q2} = \left[E''_q - E'_q + I_d\left(\frac{X'_d - X''_d}{1 + S_d}\right)\right]\left(\frac{X_d - X''_d}{X'_d - X''_d}\right) \quad (12)$$

$$E_{d1} = E''_d - E_{d2} - I_q\left(\frac{X_q - X''_q}{1 + S_q}\right) \quad (13)$$

$$E_{d2} = \left[E''_d - E'_d - I_q\left(\frac{X'_q - X''_q}{1 + S_q}\right)\right]\left(\frac{X_q - X''_q}{X'_q - X''_q}\right) \quad (14)$$

$$E_l = \sqrt{(V_q + I_q r_a + I_d X_l)^2 + (V_d + I_d r_a - I_q X_l)^2} \quad (15)$$

$$Sd = f_s(E_l) \text{ and } Sq = \frac{X_q}{X_d} f_s(E_l) \quad (16)$$

where *fs()* is the quadratic open-circuit saturation function.

The *gentpj* model is identical to the above, except that the saturation functions (*Sd* and *Sq*) are given by equation (4), in the main text of the paper, which introduces one additional parameter K_{is}, which is a scalar multiplier of total stator current.

To convert from *genrou* to *gentpf*, one only needs to switch between the models, all parameters translate one-to-one. The same is true going from *genrou* to *gentpj*. However, for *gentpj* the K_{is} parameter is needed, which can be estimated from on-line V-curve data. To go from *gentpj* to *gensal*, simply set *X'q* equal to *Xq* and *T'qo* to an extremely large value (e.g. 9999), and all other parameter translate one-to-one. By setting *T'qo* to an extremely large value we are trying to eliminate equation (5) above and remove the effect of E'q. Note: several commercial software platforms have already implemented this new (*gentpj*) model, and in those platforms to eliminate this equation, the user is asked to set *T'qo* equal to zero.

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9. Biographies

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